AI-Powered Biomedical Dataset Discovery: Bridging Search Gaps in the NCBI Gene Expression Omnibus (GEO)

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INTRODUCTION

- The NCBI Gene Expression Omnibus (GEO) contains over one million samples across ~200,000 studies, a vast but underutilized resource for biomedical discovery
- Each study includes a gene-expression matrix and accompanying textual metadata.
- Metadata quality is inconsistent, with variable terminology and incomplete annotation (e.g., "liver cancer" vs. "hepatoma"), making systematic retrieval and meta-analysis difficult.

The objective is to develop a scalable AI-driven framework for metadata curation and study retrieval in the GEO repository that integrates **Retrieval-Augmented Generation (RAG)** with **MeSH-based semantic and keyword search**, enabling accurate discovery, normalization, and categorization of studies despite the inconsistent metadata.

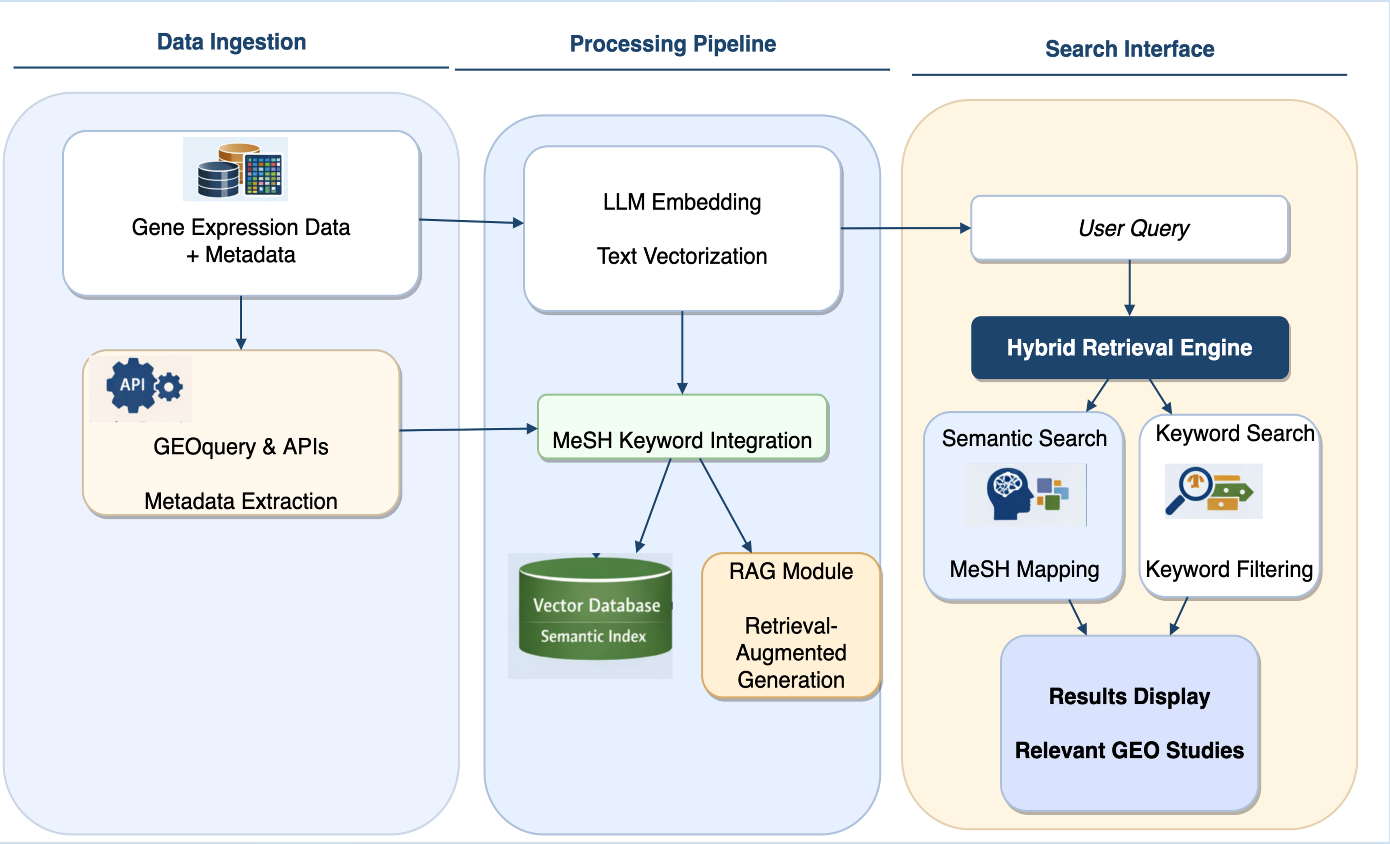
OBJECTIVES

The aim of this study:

- Dense Semantic Search
    - Milvus Vector Database (Cosine similarity)
    - Captures conceptual relationships
- MeSH (Medical Subject Headings) ontology expansion
    - MeSH 2026 - 31,000+ descriptors
    - Query expansion at search time
    - Confidence-scored dataset tagging at ingest time
- Keyword matching
    - Metadata curation and keyword filtration for better performance
- Integrate with iDEP:
    - Feed the data to the applications like iDEP, for performant semantic searches that facilitates the further analysis of the GEO Data.

MATERIALS AND METHODS

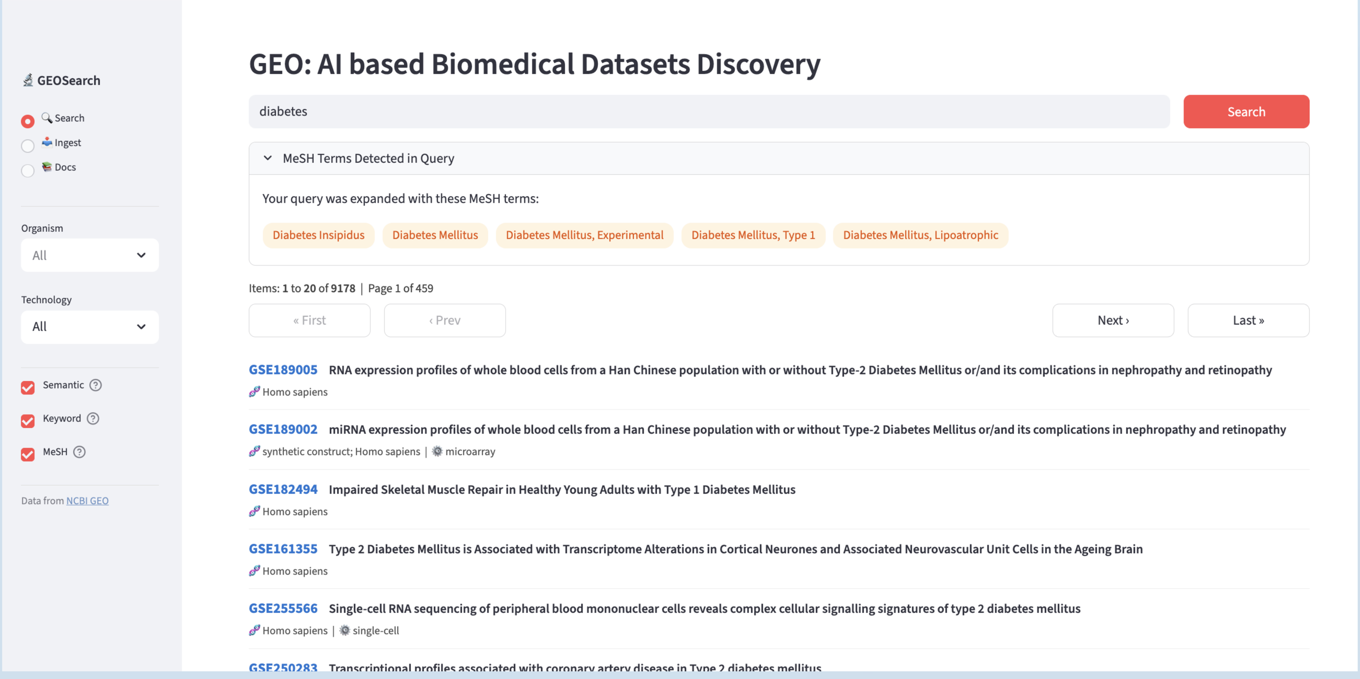
- **PostGreSQL**: GSE Series metadata storage , MeSH Terminology dictionary using PostgreSQL
- **Embedding**: Pluggable providers with OpenAI
- **Vector Storage**: **Milvus** for semantic embeddings
- **Hydrid Search Engine** : Multi-field dense search
- **API**: FastAPI with health checks and endpoints

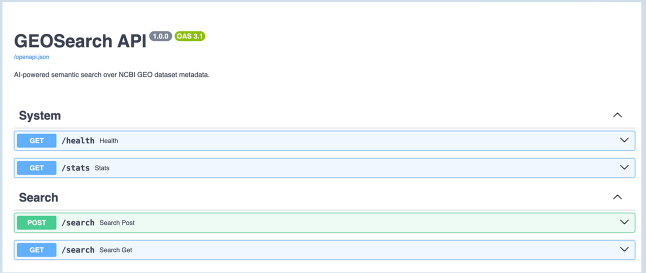


Component	Technology	Purpose
Language	Python 3.11+	Main implementation
UI Framework	Streamlit	Interactive web interface
Database	PostgreSQL 15	Structured metadata
Vector DB	Milvus 2.3.3	Semantic embeddings
ORM	SQLAlchemy 2.0	Database access
Embeddings (Testing)	sentence-transformers	Local embedding generation
Embeddings (Cloud)	OpenAI API	Cloud embedding generation
Data Source	NCBI E-utilities	GEO data
Containerization	Docker Compose	Service orchestration

RESULTS AND IMPLICATIONS

- **Muliti Layered Architecture :** Data imports from NCBI GEO using API, MeSH Terminology dictionary in PostgreSQL, create Milvus semantic embeddings
- **Integration**: Integration pathway into iDEP using API endpoints
- **Data scope**: Homosapien Datasets imports from 2000 to 2026 from NCBI GEO
- Query Expansion with MeSH terms such as
  "heart attack" → Myocardial Infarction, Coronary Disease, Heart Failure, Cardiac Remodeling
  "breast cancer"→Breast Neoplasms, BRCA2 Protein, Triple Negative Breast Neoplasms
  "diabetes" → Diabetes Mellitus Type 2, Gestational Diabetes, Insulin Resistance





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